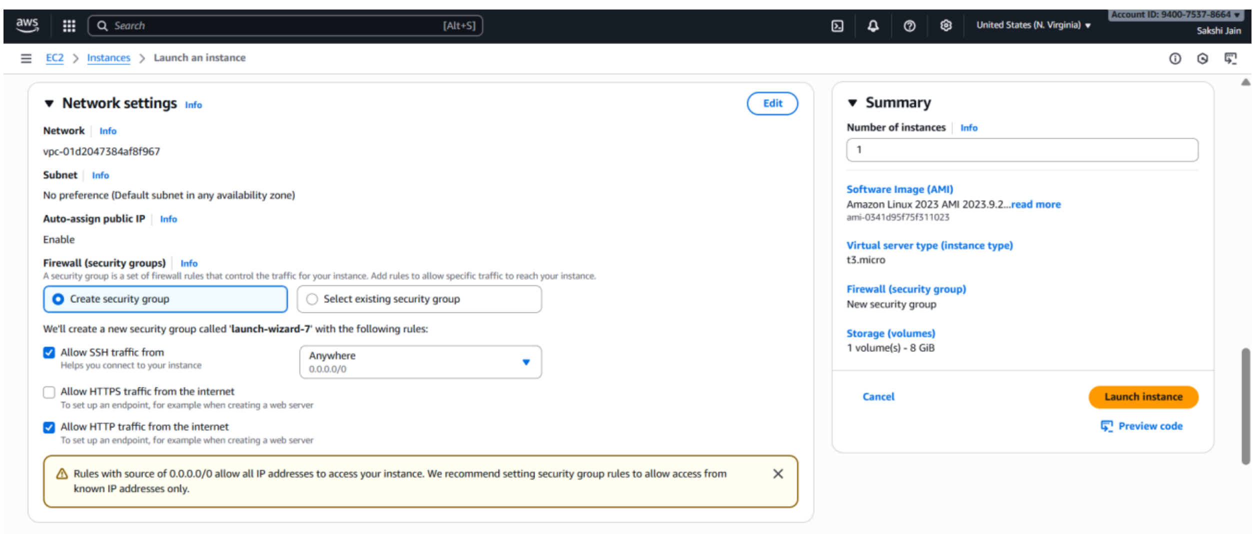
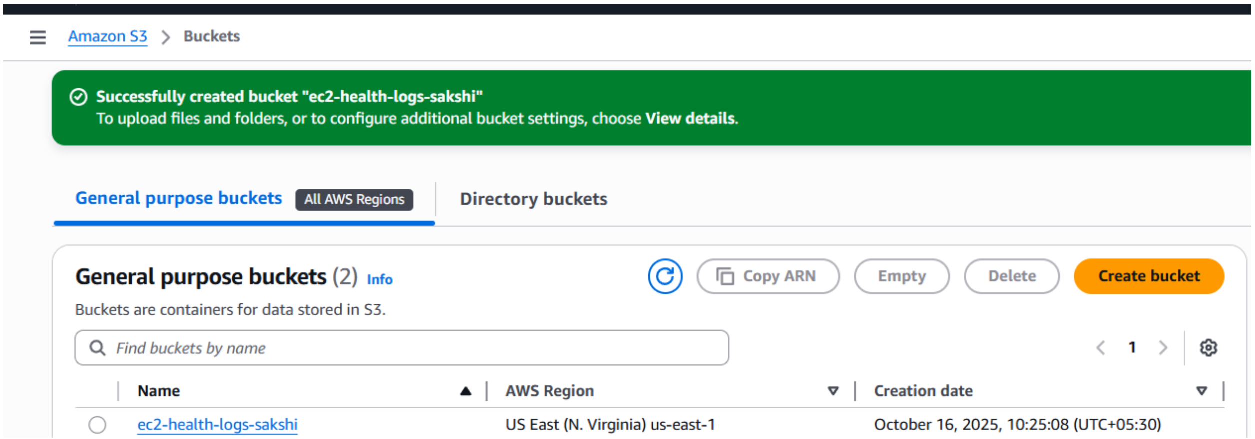
STEP 1: Create an EC2 Instance

1. Login to AWS Console → https://aws.amazon.com/console
2. Search for "EC2" → open it.
3. Click Launch Instance
   - Name: HealthCheckInstance
   - AMI: Amazon Linux 2
   - Instance type: t2.micro (Free Tier)
4. Key Pair → create a new one → download .pem file.
5. Network Settings:
   - Allow SSH (port 22)
   - Allow HTTP (port 80)
6. Click Launch Instance ✅



STEP 2: Create an S3 Bucket for Logs

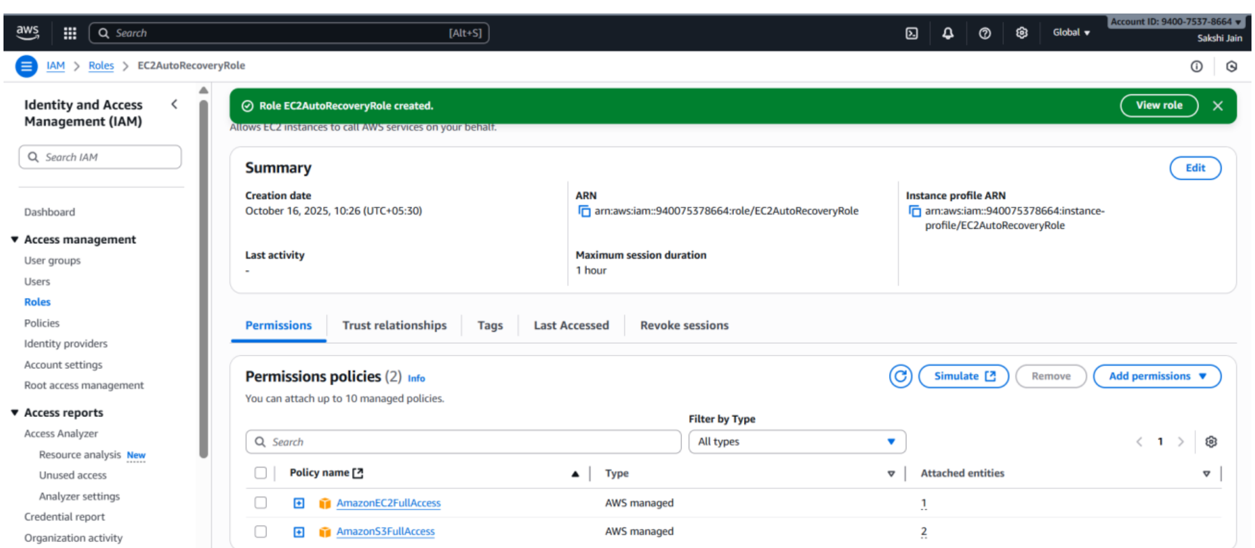
1. Go to S3 → Create bucket
   - Name: ec2-health-logs-sakshi
   - Region: same as your EC2 region
2. Keep all defaults → Create bucket

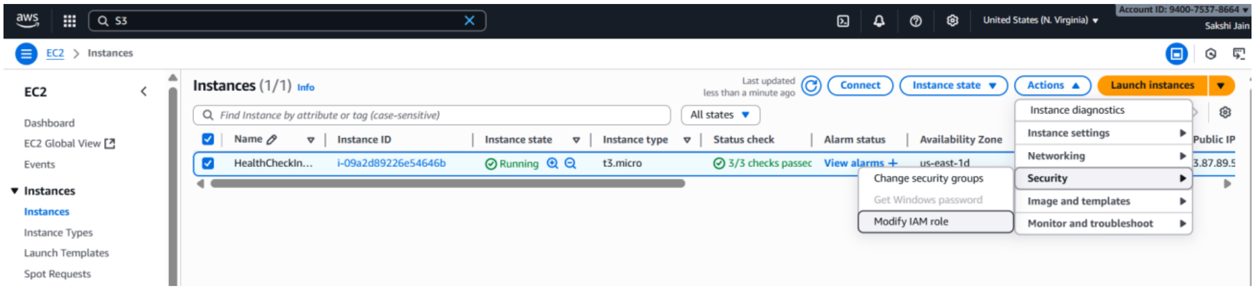


STEP 3: Create IAM Role for EC2

This gives EC2 permission to restart itself & write logs to S3.

1. Go to IAM → Roles → Create role
2. Choose AWS Service → EC2
3. Click Next and attach these policies:
    - AmazonS3FullAccess
    - AmazonEC2FullAccess
4. Name the role: EC2AutoRecoveryRole
5. Click Create role
6. Go to EC2 → Instances → Select your instance → Actions → Security → Modify IAM Role
    - Attach the role EC2AutoRecoveryRole ✅

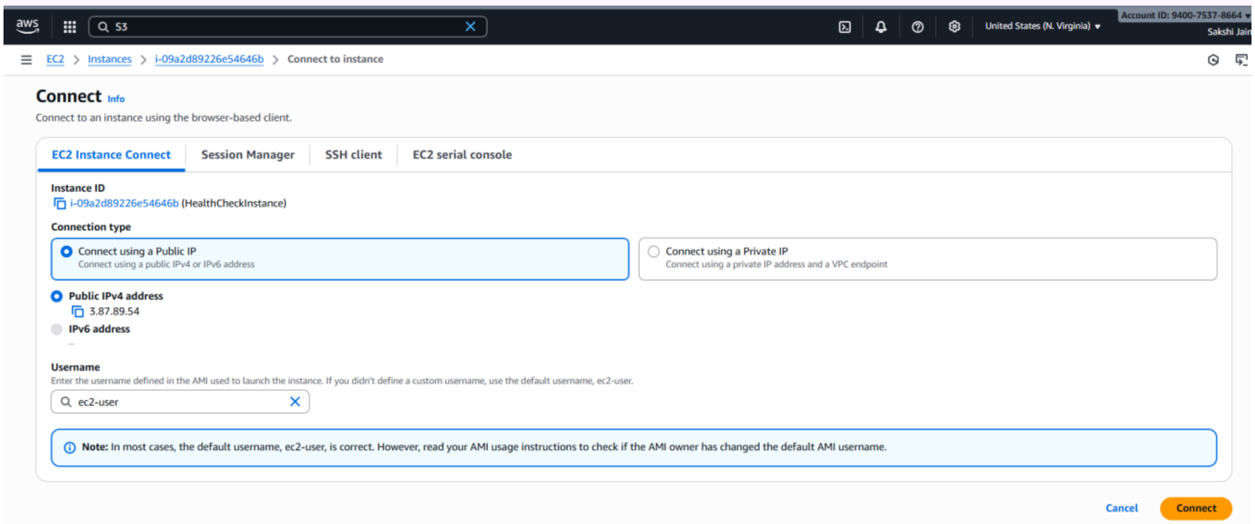




STEP 4: Connect to EC2 Instance

From your terminal / command prompt:

ssh -i "your-key.pem" ec2-user@<EC2-Public-IP>



STEP 5: Install AWS CLI (if not already)

sudo yum update -y
sudo yum install aws-cli -y

Check:

aws --version

STEP 6: Create the Health Check Script

Inside EC2 terminal:

vim healthcheck.sh

Paste this script 👇

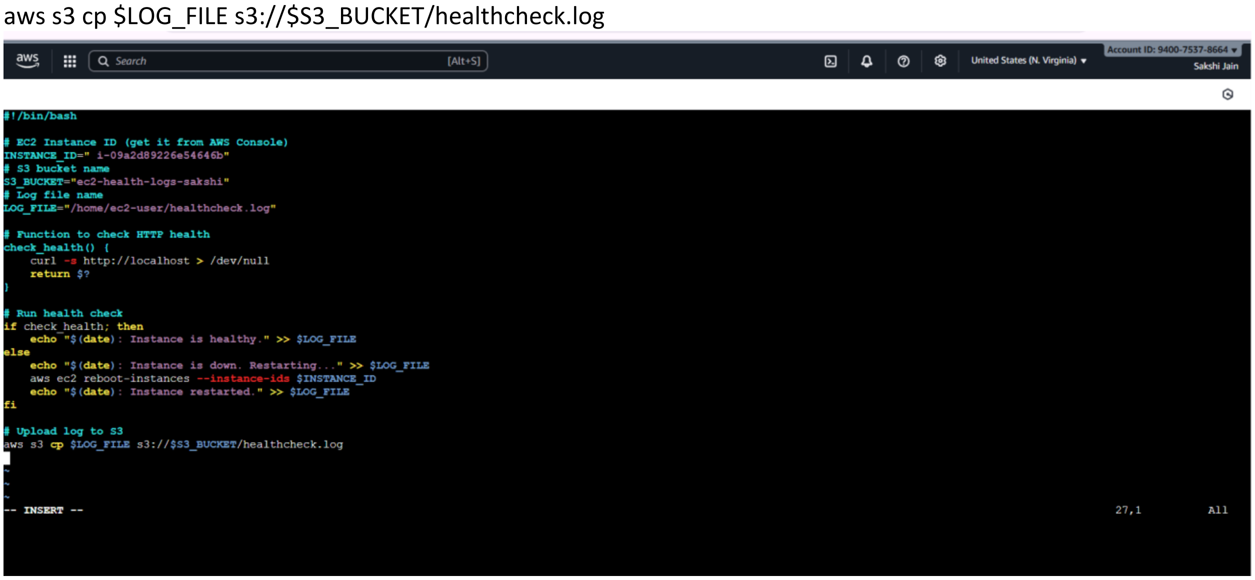
```
#!/bin/bash

# EC2 Instance ID (get it from AWS Console)
INSTANCE_ID=" i-09a2d89226e54646b"
# S3 bucket name
S3_BUCKET="ec2-health-logs-sakshi"
# Log file name
LOG_FILE="/home/ec2-user/healthcheck.log"

# Function to check HTTP health
check_health() {
   curl -s http://localhost > /dev/null
   return $?
}

# Run health check
if check_health; then
   echo "$(date): Instance is healthy." >> $LOG_FILE
else
   echo "$(date): Instance is down. Restarting..." >> $LOG_FILE
   aws ec2 reboot-instances --instance-ids $INSTANCE_ID
   echo "$(date): Instance restarted." >> $LOG_FILE
fi

# Upload log to S3
aws s3 cp $LOG_FILE s3://$S3_BUCKET/healthcheck.log
```



STEP 7: Make the Script Executable

chmod +x healthcheck.sh

STEP 8: Schedule the Script Automatically (Cron Job)

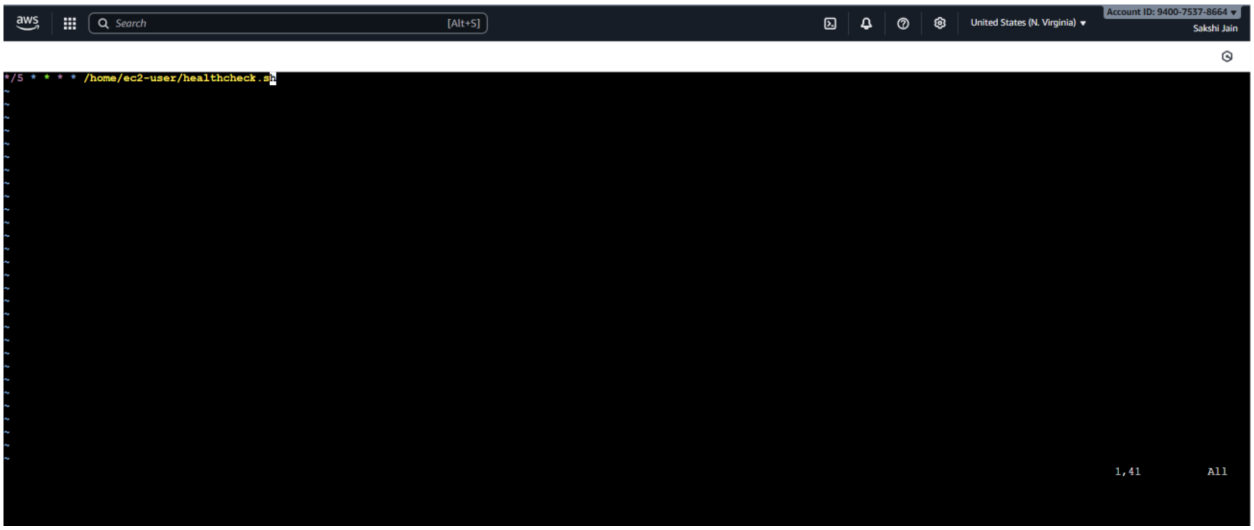
Open crontab:

crontab -e

Add this line (run every 5 minutes):

*/5 * * * * /home/ec2-user/healthcheck.sh

Save and exit.



STEP 9: Testing (How to Check Final Output)

Test 1: Normal Condition

Run manually first: Run with Root user and Need to Start and Stop HTTPD service first.

./healthcheck.sh

Then check: from Ec2 user(local User)

cat healthcheck.log

You should see:

Thu Oct 16 10:00:00 UTC 2025: Instance is healthy.
Also check your S3 bucket — it will have healthcheck.log.

Test 2: Simulate "Instance Down"

To test recovery:

1. Stop your web service (simulate failure):
2. sudo systemctl stop httpd
3. Run script again:
4. ./healthcheck.sh
5. It will detect "down", restart EC2, and log this:
6. Thu Oct 16 10:05:00 UTC 2025: Instance is down. Restarting...
7. Thu Oct 16 10:05:15 UTC 2025: Instance restarted.
8. Check S3 bucket again → you'll see updated logs uploaded automatically.

FINAL OUTPUT (Summary)

EC2 instance automatically checked
Automatically restarted when down
Logs stored to S3
Continuous monitoring via cron